```
!pip install -q transformers sentencepiece torch
```

```
from transformers import pipeline

print("Import successful")
```

```
Import successful
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Due to an internal KeyError in the transformers pipeline for translation tasks,
# we are manually loading the model and tokenizer to create a translation function.

model_name = "Helsinki-NLP/opus-mt-en-fr"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Define a function that mimics the behavior of a translation pipeline
def translate_en_to_fr(text):
    inputs = tokenizer(text, return_tensors="pt", padding=True, truncation=True)
    # The generate method returns sequences of token ids
    outputs = model.generate(**inputs)
    # Decode the generated token ids back to text
    return tokenizer.decode(outputs[0], skip_special_tokens=True)

# Assign the translation function to the variable name the user intended
translator_fr = translate_en_to_fr

print(f"Translator for {model_name} initialized successfully.")
```

```
tokenizer_config.json: 100%   42.0/42.0 [00:00<00:00, 1.12kB/s]
source.spm: 100%   778k/778k [00:00<00:00, 11.9MB/s]
target.spm: 100%   802k/802k [00:00<00:00, 36.6MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downl
vocab.json:   1.34M/? [00:00<00:00, 33.9MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: 100%   301M/301M [00:02<00:00, 178MB/s]
Loading weights: 100%   258/258 [00:00<00:00, 401.94it/s, Materializing param=model.shared.weight]
model.safetensors: 100%   301M/301M [00:03<00:00, 149MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s
generation_config.json: 100%   293/293 [00:00<00:00, 11.7kB/s]
Translator for Helsinki-NLP/opus-mt-en-fr initialized successfully.
```

```
print(translator_fr("I love artificial intelligence"))
```

```
J'aime l'intelligence artificielle
```

```python
# Manual setup for English to German translation (due to pipeline KeyError)
model_name_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

def translate_en_to_de(text):
    inputs = tokenizer_de(text, return_tensors="pt", padding=True, truncation=True)
    outputs = model_de.generate(**inputs)
    return tokenizer_de.decode(outputs[0], skip_special_tokens=True)

translator_de = translate_en_to_de
print(f"Translator for {model_name_de} initialized successfully.")

# Manual setup for English to Hindi translation (due to pipeline KeyError)
model_name_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_hi = AutoTokenizer.from_pretrained(model_name_hi)
model_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_hi)

def translate_en_to_hi(text):
    inputs = tokenizer_hi(text, return_tensors="pt", padding=True, truncation=True)
    outputs = model_hi.generate(**inputs)
    return tokenizer_hi.decode(outputs[0], skip_special_tokens=True)

translator_hi = translate_en_to_hi
print(f"Translator for {model_name_hi} initialized successfully.")
```

```
config.json:   1.33k/? [00:00<00:00, 19.2kB/s]
tokenizer_config.json: 100%   42.0/42.0 [00:00<00:00, 938B/s]
source.spm: 100%   768k/768k [00:00<00:00, 7.34MB/s]
target.spm: 100%   797k/797k [00:00<00:00, 11.7MB/s]
vocab.json:   1.27M/? [00:00<00:00, 7.41MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: 100%   298M/298M [00:03<00:00, 130MB/s]
model.safetensors: 100%   298M/298M [00:05<00:00, 76.6MB/s]
Loading weights: 100%   258/258 [00:00<00:00, 549.17it/s, Materializing param=model.shared.weight]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s
generation_config.json: 100%   293/293 [00:00<00:00, 23.1kB/s]
Translator for Helsinki-NLP/opus-mt-en-de initialized successfully.
config.json:   1.39k/? [00:00<00:00, 60.7kB/s]
tokenizer_config.json: 100%   44.0/44.0 [00:00<00:00, 1.70kB/s]
source.spm: 100%   812k/812k [00:00<00:00, 3.52MB/s]
target.spm: 100%   1.07M/1.07M [00:00<00:00, 2.82MB/s]
vocab.json:   2.10M/? [00:00<00:00, 7.66MB/s]
pytorch_model.bin: 100%   306M/306M [00:08<00:00, 34.3MB/s]
Loading weights: 100%   258/258 [00:00<00:00, 432.01it/s, Materializing param=model.shared.weight]
model.safetensors: 100%   306M/306M [00:03<00:00, 146MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, s
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, s
generation_config.json: 100%   293/293 [00:00<00:00, 12.2kB/s]
Translator for Helsinki-NLP/opus-mt-en-hi initialized successfully.
```

```python
sentences = [
    "I love learning new technologies.",
    "The weather is very pleasant today.",
    "Artificial Intelligence is transforming industries.",
    "Can you help me with this problem?",
    "She went to the market and bought fruits.",
    "Although it was raining, we continued the match.",
    "Data science requires statistics and programming knowledge.",
    "What are you doing right now?",
    "He completed the task before the deadline.",
    "Social media influences public opinion.",
    "Machine learning models require proper training.",
    "The results were surprising and unexpected.",
    "Do you understand this concept clearly?",
    "The system failed due to a technical error.",
```

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    "We should work together to solve this issue."
]
```

```
results = []

for sentence in sentences:
    fr = translator_fr(sentence)
    de = translator_de(sentence)
    hi = translator_hi(sentence)

    results.append((sentence, fr, de, hi))
```

```
for res in results:
    print("\nOriginal:", res[0])
    print("French:", res[1])
    print("German:", res[2])
    print("Hindi:", res[3])
```

```
French: Pouvez-vous m'aider avec ce problème ?
German: Können Sie mir bei diesem Problem helfen?
Hindi: आप इस समस्या के साथ मेरी मदद कर सकते हैं?

Original: She went to the market and bought fruits.
French: Elle est allée au marché et a acheté des fruits.
German: Sie ging auf den Markt und kaufte Früchte.
Hindi: वह बाजार में गया और फल लाया.

Original: Although it was raining, we continued the match.
French: Même s'il pleuvait, nous avons continué le match.
German: Obwohl es regnete, setzten wir das Spiel fort.
Hindi: हालाँकि बारिश हो रही थी, फिर भी हमने मैच जारी रखा ।

Original: Data science requires statistics and programming knowledge.
French: La science des données exige des statistiques et des connaissances en matière de programmation.
German: Die Datenwissenschaft erfordert Statistiken und Programmierkenntnisse.
Hindi: डाटा विज्ञान के लिए सांख्यिकी और प्रोग्रामिंग ज्ञान की जरूरत है.

Original: What are you doing right now?
French: Qu'est-ce que tu fais en ce moment ?
German: Was machst du gerade?
Hindi: तुम अभी क्या कर रहे हो?

French: Les résultats ont été surprenants et inattendus
```

```
French: Les résultats ont été surprenants et inattendus.
German: Die Ergebnisse waren überraschend und unerwartet.
Hindi: परिणाम आश्चर्यचकित और अप्रत्याशित थे ।

Original: Do you understand this concept clearly?
French: Comprenez-vous clairement ce concept ?
German: Verstehen Sie dieses Konzept klar?
Hindi: क्या आप इस धारणा को स्पष्ट रूप से समझते हैं?

Original: The system failed due to a technical error.
French: Le système a échoué en raison d'une erreur technique.
German: Das System scheiterte aufgrund eines technischen Fehlers.
Hindi: तकनीकी त्रुटि के कारण तंत्र असफल.

Original: We should work together to solve this issue.
French: Nous devrions travailler ensemble pour résoudre ce problème.
German: Wir sollten zusammenarbeiten, um dieses Problem zu lösen.
Hindi: इस मसले को सुलझाने के लिए हमें साथ मिलकर काम करना चाहिए ।
```

```
import pandas as pd

df = pd.DataFrame(results, columns=["English", "French", "German", "Hindi"])
```